A Project Report on

AI DRIVEN AUTONOMOUS STEERING USING IoT AND ML

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

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Abstract

Autonomous vehicle technology represents a major leap in modern transportation, offering the potential for safer, more efficient, and sustainable mobility. However, its adoption is limited by the high cost and complexity of current systems, which often depend on expensive sensors, advanced processors, and proprietary architectures. To address these challenges, this project presents a low-cost autonomous steering control system designed to make self-driving research more accessible to students and small-scale developers. The system employs a Convolutional Neural Network (CNN) to predict steering angles from real-time camera input, which are then physically actuated through an Arduino-controlled stepper motor and steering mechanism. By using affordable components instead of high-end sensors like LiDAR or RADAR, the project demonstrates that reliable autonomous steering can be achieved with minimal hardware investment. This approach not only reduces development costs but also encourages open innovation and experimentation in the field of intelligent transportation. The proposed design serves as a scalable foundation for future enhancements such as braking, acceleration, and object detection, ultimately contributing to the development of more inclusive, sustainable, and cost-efficient autonomous vehicle technologies.

This project aligns with several United Nations Sustainable Development Goals (SDGs). It directly supports SDG 9: Industry, Innovation, and Infrastructure by promoting low-cost and scalable innovation in autonomous vehicle technology, making advanced transportation research accessible to students and small-scale developers. In line with SDG 11: Sustainable Cities and Communities, the project contributes to the development of smart and efficient mobility solutions that can enhance urban transport systems and reduce congestion. Additionally, it addresses SDG 13: Climate Action by emphasizing energy-efficient and eco-friendly design principles, ultimately aiming to lower the carbon footprint of future autonomous mobility systems.

Keywords: Autonomous steering, Internet of Things (IoT), Machine Learning (ML), Arduino, Stepper motor, Steering angle prediction, Intelligent vehicles, Low-cost automation

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ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
RL	Reinforcement Learning
PPO	Proximal Policy Optimization
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
GRU	Gated Recurrent Unit
DQN	Deep Q-Network
MDP	Markov Decision Process
NTL	Non-Technical Losses
SAIDI	System Average Interruption Duration Index
SDG	Sustainable Development Goals
GAN	Generative Adversarial Network
CTGAN	Conditional Tabular Generative Adversarial Network
AutoML	Automated Machine Learning
PR-AUC	Precision–Recall Area Under the Curve
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
IoT	Internet of Things
BigQuery	Google BigQuery (cloud data warehouse)

Chapter 1

Introduction

1. Introduction

Autonomous vehicles aim to improve road safety and driving efficiency by reducing dependence on human control. Among all autonomous functions, steering is the most critical, as it directly governs lateral motion, lane keeping, and trajectory tracking. A major breakthrough in this domain was achieved by Bojarski et al. [1], who demonstrated an end-to-end deep learning framework that maps raw camera images directly to steering commands. This work eliminated the need for handcrafted features and explicit vehicle models. Building on this idea, Chen et al. [2] introduced a direct perception approach that predicts intermediate driving affordances, creating a structured bridge between vision and control.

To improve temporal consistency and robustness, Valiente et al. [3] proposed a CNN–LSTM architecture for steering angle control, while Yang et al. [4] incorporated multi-modal inputs such as visual and vehicle state data. Further refinement of CNN-based regression models for continuous steering prediction was presented by Aires Da Conceicao and Degadwala [5], and more recent deep learning frameworks confirm the feasibility of accurate steering angle prediction in real-world scenarios [6]. These studies collectively establish machine learning as a powerful alternative to traditional steering control.

Conventional steering systems, however, have long relied on classical and model-based control methods such as nonlinear control and predictive control [7], [8]. With the rise of electric power steering and autonomous electric vehicles, researchers have developed ANN-based EPS systems [9], distributed drive steering architectures [10], and trajectory control strategies for in-wheel motor vehicles [11], [12]. Predictive active steering using model predictive control has further improved stability and performance in challenging driving conditions [13], [14]. Fault-tolerant steering mechanisms [15] and performance assessment of EPS systems [16] ensure reliability under real-world uncertainties.At the system level, complete autonomous vehicle implementations [17], ROS-based drive-by-wire architectures [18], and early path-following platforms [19] demonstrate how perception, planning, and actuation are integrated in real time vehicles are an emerging technology designed to operate without human intervention by using advanced sensing, computation, and control systems. These vehicles aim to enhance road safety, reduce traffic congestion, and improve driving efficiency. They rely on a combination of sensors such as cameras, GPS, and inertial measurement units (IMUs) to perceive the environment and make driving decisions.

Modern autonomous systems integrate perception, planning, and control modules to enable safe navigation. Among these, steering coAutonomountrol is one of the most critical components, as it directly affects the vehicle's ability to follow a path and maintain lane discipline. With the advancement of artificial intelligence and embedded systems, autonomous vehicles are becoming more reliable and capable of handling real-world driving scenarios.

Steering angle prediction is a key task in autonomous driving, responsible for determining how much the vehicle should turn to follow a desired trajectory. It plays a crucial role in lane keeping, path tracking, and obstacle avoidance. Traditional steering methods based on control theory, such as PID and model-based controllers, depend heavily on predefined mathematical models and often fail in dynamic environments.

Recent advancements in machine learning have introduced data-driven approaches for steering control. Techniques such as convolutional neural networks (CNNs), regression models, and reinforcement learning enable systems to learn directly from driving data. End-to-end learning models can map raw sensor inputs, especially camera images, directly to steering commands, eliminating the need for manual feature extraction. These methods provide improved adaptability and performance in complex driving conditions

The integration of Internet of Things (IoT) technologies has significantly enhanced the capabilities of autonomous steering systems. IoT enables continuous communication between sensors, processing units, and actuators, forming a real-time decision-making framework. Sensors such as cameras capture visual information, IMUs provide motion data, and GPS ensures accurate positioning.This sensor data is processed by embedded systems to predict steering angles and control vehicle motion. IoT-based architectures allow seamless data flow and synchronization, improving system responsiveness and reliability. Sensor fusion techniques further enhance accuracy by combining multiple data sources, reducing the impact of noise and environmental uncertainties

The increasing complexity of driving environments highlights the need for intelligent and adaptive steering systems. Traditional approaches are limited in handling dynamic scenarios, making machine learning-based solutions essential. This study focuses on analyzing various steering angle prediction techniques using machine learning and IoT technologies.

The scope of this work includes reviewing vision-based, sensor-based, and hybrid approaches, along with evaluating different models and performance metrics. It also identifies key challenges such as real-time processing, generalization, and safety constraints, while suggesting future directions for developing efficient and cost-effective autonomous steering systems.

Chapter 2

Literature Survey/ Existing System

2. Survey/ Existing System

[1] Golluri Ramcharan and Dhunnapothula Raghu, "Arduino based stepper motor speed regulation for robotics applications," 2024

This paper presents a stepper motor drive using the hybrid two-phase model. The stepper motor changes the pulse signals into angular displacement with some angles. Stepper motors are used to control the speed and are also more reliable for smooth operations. The stepper motor provides constant holding torque with controlled speed ranges from 0 to 6000 range. The closed-loop control technique with park transformation is used to control the speed torque variables in a design range. The simulation and hardware results were discussed in MATLAB\Simulink software. The verified simulation results are motor source voltage, motor source current, motor speed, and torque. The hardware results are also implemented in this paper, the implemented circuit by using Arduino microcontroller.

[2] Dr. Ruchi Pandey, Anand Goswami, Uttam Lal Yadav, and Yogesh Vishwakarma, "Designing and Crafting Stepper Motor Position Control with Arduino Mega 2560," Feb. 2023, International Journal Of Innovation In Engineering Research and Management)

This paper describes the design of a position-control system for stepper motors using the Arduino Mega 2560 microcontroller. It implements closed-loop control by integrating rotary encoders as feedback devices to precisely monitor and correct motor position errors. The system can independently or synchronously control multiple motors with high resolution, using the Mega's many I/O and PWM capabilities. The approach aims for smooth rotation under physical constraints and improved accuracy over open-loop systems by reducing step loss or slip

[3] Amad Narto, Suharto Suharto, and Regga Diko Catur, "The Prototype Electric Steering Gear Based Microcontroller Arduino," 2023

This work describes the design and implementation of a demonstrative electric steering gear system using an Arduino-based microcontroller setup. The authors employ a Research & Development methodology to iteratively build and refine a prototype comprising three main modules: the ESP8266 microcontroller, a servo motor for steering actuation, and a Wi-Fi communication module for remote control. Users can adjust steering angle positions via a smartphone interface, with the system responding in real time by moving the steering shaft accordingly. The design is intended as a teaching and demonstration tool, replacing conventional mechanical control levers with smart, networked actuation, and facilitating more precise and convenient control of auxiliary machinery like ship steering systems.

[4] ARM Khairudin, "Design of Stepper Motor Application as Archery Drawer," 2023

This study presents the design and implementation of an automated archery drawer system utilizing a stepper motor, Arduino microcontroller, and servo motor to simulate the draw and release actions of archery. The system allows users to set the draw length between 12 to 24 cm, with the optimal performance achieved at a 22 cm draw length, resulting in a shooting efficiency of 96.7%. The project aims to provide a practical training tool for archers, enabling them to understand the impact of draw length on arrow power and accuracy, thereby enhancing their skills and reducing the risk of injury associated with prolonged training sessions.

[5] M. S. Mohammed et al., "Low-cost autonomous car level 2: Design and

implementation," 2023

Modern cars use autonomous systems to enhance driving, but most conventional vehicles lack Driving Assist Systems (DAS), which are costly and complex to implement due to ECU security. This paper presents an intelligent DAS for real-time steering angle prediction using deep learning (DL) with data collected from real environments. An object detection model warns the driver of obstacles and measures distances. DL outputs feed into an Electronic Power Steering (EPS) system, where the steering angle is measured via a sensor and adjusted automatically. Real-time tests on a 2009 Toyota Corolla using a camera, CAN-BUS messages, and a steering sensor demonstrate effective intelligent steering control and driver assistance.

[6] Development of Four-Wheel Steering (4WS) with Arduino-Controlled Stepper Motor," Nov. 2022

This project aims to develop an alternative to traditional two-wheel steering systems to improve vehicle maneuverability and handling. Conventional steering mechanisms like the Ackerman or Davis systems have limitations in achieving minimal turning radii. To address this, a four-wheel steering (4WS) system is proposed, enabling vehicles to execute turns with a reduced turning radius around their center of gravity. The system incorporates a rack and pinion steering linkage at the rear, controlled by an Arduino Uno microcontroller equipped with a stepper motor. This design allows for enhanced control and maneuverability, particularly in navigating narrow roads or parking spaces. The project provides a comprehensive understanding of 4WS technology, exploring its advantages over traditional systems and its potential impact on the automotive industry.

[7] "Development of Steering Control System for a Small Electric Vehicle," Nov. 2022, YMER Digital

This paper presents the development of a steering control system for a small electric vehicle, focusing on the retrofitting of its mechanical steering to an electric power-assisted system. The existing steering mechanism is replaced with a closed-loop stepper motor, specifically a NEMA 23 standard closed-loop stepper motor with a gearbox, controlled via an Arduino microcontroller. The system utilizes onboard 48V DC power to actuate the motor, which is directly coupled to the steering column through specially designed gears. The control algorithm employs the Pure Pursuit path tracking method, with lane data input derived from

image processing techniques. The vehicle's performance is evaluated at various speeds (10 km/h, 15 km/h, 20 km/h, and 25 km/h), demonstrating the system's capability to enhance steering precision and assistive control in real-time driving scenarios.

[8] "Electronic Hardware Design of An Autonomous Vehicle," 2020, IJERT

This paper presents the design of electronic hardware for an autonomous vehicle capable of real-time decision-making without human intervention. The system integrates sensors to capture environmental data, which is processed using algorithms involving machine learning, artificial intelligence, neural networks, and image processing. The processed data informs the steering, braking, and acceleration mechanisms, enabling the vehicle to navigate safely and efficiently. The proposed hardware aims to enhance the reliability and safety of autonomous vehicles by providing accurate outputs in real-time applications.

[9] A. Rossi, N. Ahmed, S. Salehin, T. Hasnine Choudhury, and G. Sarowar, "Real-time Lane Detection and Motion Planning in Raspberry Pi and Arduino for an Autonomous Vehicle Prototype," Sep. 20, 2020

This paper presents an autonomous vehicle prototype capable of lane detection and motion planning without human intervention. The system utilizes a Pi Camera 1.3 to capture real-time video, which is processed by a Raspberry Pi 3.0 Model B using Python 3.7.4 and OpenCV 4.2. Lane detection is achieved through the Canny edge detection algorithm and Hough transformation, while lane tracking employs the Kalman filter prediction method. The Arduino Uno controls the motor via a PID algorithm, adjusting steering based on the detected lane's midpoint. The steering direction is further refined using the Past Accumulation Average Method and Kalman Filter Prediction Method. Real-time testing in a controlled environment demonstrates the system's effectiveness in lane detection and autonomous motion planning.

[10] R. Thrun et al., "Stanley: The Robot that Won the DARPA Grand Challenge," 2006, Journal of Field Robotics

This paper details the development and performance of Stanley, an autonomous vehicle that won the 2005 DARPA Grand Challenge by completing a 132-mile desert course in under seven hours. Developed by the Stanford Racing Team, Stanley was equipped with a suite of sensors, including LIDAR, radar, GPS, and cameras, to perceive its environment. Its software

architecture integrated machine learning and probabilistic reasoning to enable real-time decision-making and navigation. The vehicle's success demonstrated the viability of autonomous driving technologies in complex, unstructured environments. Stanley's performance marked a significant milestone in robotics and paved the way for advancements in autonomous vehicle research.

Table 1: Research Gap

Year	Title	Domain	Algorithm	Dataset
2024	Arduino based stepper motor speed regulation for robotics applications (Ramcharan & Raghu)	Robotics / Motor control / Power electronics	Closed-loop control using Park Transformation with hybrid two-phase model; simulation in MATLAB/Simulink + hardware with Arduino.	Experimental & simulation data: motor source voltage, current, speed, torque; hardware implementation. Speed range: 0 to ~6000 (units presumably pulses or rpm)
2023	Designing and Crafting Stepper Motor Position Control with Arduino Mega 2560	Robotics / Position control / Embedded Control	Position control using Arduino Mega 2560 + Rotary Encoder feedback; likely closed-loop control.	Experimental: synchronous control of two stepper motors; using position feedback via encoder; testing for accuracy in position control.
2023	The Prototype Electric Steering Gear Based	Remote / auxiliary steering gear demo; maritime ships auxiliary machinery control	Research & Development procedural model; uses ESP8266	Hardware prototype: stepper motor + driver; experimental

	Microcontroller Arduino		module for WiFi, Servo Motor for steering direction/angle; control via smartphone app; microcontroller logic.	motion of archery drawer; no standardized dataset; measurements are likely mechanical movement, angle, perhaps tension.
2023	Low cost autonomous car level 2: Design and implementation	Autonomous vehicles / driver assistance / automation	Probably uses sensor fusion, planning & control methods; but specific algorithm details not found in summary. Could include vision, steering control, path planning.	Experimental prototype; sensors, hardware; no public large dataset info; likely own collected test trials.
2022 / 2023	Development of Four Wheel Steering (4WS) with Arduino Controlled Stepper Motor	Vehicle steering systems / robotics / embedded control	Use of stepper motors to achieve four wheel steering; Arduino for control; details of steering angle mapping etc.	Prototype hardware, controlled experiments/bench tests: steering behaviour etc; no public large dataset.
2022	Development of Steering Control System for a Small Electric Vehicle	EV control / steering systems	Likely geometry based / feedback control of steering; possibly PID or simpler control laws.	Hardware setup; small EV; experimental testing; again not large benchmark dataset.

Chapter 3

Limitation of Existing System

3.Limitation of Existing System

Existing autonomous steering systems face several significant challenges that hinder their adoption, particularly in educational and research settings. One of the primary limitations is the high cost of implementation, as most systems rely on expensive components such as LiDAR sensors, high-resolution cameras, and specialized computing hardware. This makes them unaffordable for students, small research groups, and hobbyists who wish to explore autonomous navigation technologies. Moreover, these systems often involve complex hardware integration, requiring multiple sensors, microcontrollers, and control units to be connected and calibrated accurately. Such complexity increases setup time, maintenance effort, and the likelihood of system malfunction. Another major drawback is the limited accessibility of commercial systems, which are typically closed-source or proprietary, restricting opportunities for experimentation, customization, and innovation in academic environments.

Additionally, most high-end autonomous systems are designed to operate efficiently on powerful GPUs or embedded processors, while low-cost setups struggle with real-time processing and accurate steering predictions, thereby limiting their practical usability. The AI models used in these systems also face challenges such as overfitting and poor generalization, performing well only within the confines of their training data and failing under diverse real-world conditions involving varying lighting, road surfaces, or obstacles. Furthermore, many existing solutions depend on a single type of input data, such as camera feeds, without

incorporating multi-sensor fusion, which reduces their robustness under adverse environmental conditions like rain, glare, or shadows. Finally, safety and reliability concerns remain a critical limitation, as these systems often produce inconsistent results in dynamic or unpredictable environments, posing risks to both the vehicle and its surroundings.

Collectively, these challenges emphasize the urgent need for a cost-effective, modular, and IoT-integrated AI-driven autonomous steering system that can function efficiently on low-cost hardware while maintaining real-time responsiveness, adaptability, and operational safety.

Chapter 4

Problem Statement and Objective

4. Problem Statement and Objective

4.1. Problem Statement

As autonomous vehicles move closer to mainstream adoption, steering control remains one of the most critical components of self-driving technology. Accurate and responsive steering is essential for lane-keeping, obstacle avoidance, and overall vehicle stability, directly influencing passenger safety and driving efficiency. However, developing reliable autonomous steering systems often requires costly sensors, complex algorithms, and high-performance hardware, limiting access for students and small-scale innovators. This lack of affordability and openness in current research platforms creates a barrier to experimentation, learning, and innovation.

To address this gap, this project focuses on building a low-cost autonomous steering control system using Convolutional Neural Networks (CNNs) and embedded hardware to predict and actuate steering angles in real time. By simplifying hardware requirements and emphasizing practical implementation, the project aims to make autonomous steering development more accessible, educational, and scalable, laying the groundwork for future advancements in full self-driving systems.

4.2. Objectives

Low-Cost Autonomous Steering System

The primary objective of this project is to design and implement a low-cost autonomous steering control system that integrates deep learning, computer vision, and embedded hardware to mimic human driving behavior. The system aims to demonstrate how artificial intelligence can be effectively combined with IoT components to achieve real-time vehicle control without relying on expensive sensors or proprietary systems.

CNN-Based Steering Angle Prediction

A key goal is to train a Convolutional Neural Network (CNN) using video datasets converted into image frames, enabling the model to analyze road visuals and accurately predict steering angles. This allows the system to interpret real-world driving conditions and make autonomous steering decisions similar to human drivers.

Hardware Integration and Control

The project further focuses on translating CNN predictions into physical steering actions through an Arduino-controlled stepper motor connected to a steering wheel mechanism. This integration bridges the gap between software-based predictions and mechanical movement, ensuring smooth and responsive steering control.

Accuracy and Performance Optimization

To enhance system reliability, the project incorporates speed control and noise filtering mechanisms, ensuring accurate, stable, and smooth steering performance under varying conditions. These improvements contribute to the precision and safety of the autonomous control process.

Scalability and Future Expansion

The final objective is to develop a modular and scalable platform that can be extended to include additional functionalities such as accelerator and brake control, lane-keeping assistance, and obstacle detection. This ensures that the system serves as a robust foundation for further research and educational applications in autonomous driving technology.

4.3. Scope

Target Users

The AI-Driven Autonomous Steering Control System is primarily designed for students, researchers, and educators in the fields of Artificial Intelligence, Machine Learning, and Robotics. It provides an affordable and accessible platform for learning and experimentation in autonomous vehicle technology, encouraging innovation without the need for expensive hardware.

Functional Capabilities

The system utilizes a Convolutional Neural Network (CNN) trained on driving video datasets to analyze road visuals and predict steering angles in real time. These predictions are then executed through an Arduino-controlled stepper motor connected to a steering wheel, demonstrating the effective integration of deep learning with embedded hardware for practical steering automation.

System Architecture

The architecture combines computer vision, deep learning, and IoT-based control mechanisms to achieve smooth and responsive steering performance. The CNN processes video input, while the Arduino interprets and actuates the corresponding steering actions. Features such as speed control and noise filtering enhance accuracy and stability.

Scalability and Future Enhancements

Designed for modularity and scalability, the system can be extended to include accelerator and brake control, lane-keeping assistance, and obstacle detection. This flexibility makes it a strong foundation for future research and educational projects in autonomous driving and AI-based control systems.

Chapter 5

Proposed System

5. Proposed System

The proposed system presents a low-cost, AI-driven autonomous steering control framework that integrates machine learning, computer vision, and IoT-based hardware components. The system is designed to predict steering angles from real-time video input and translate those predictions into physical steering actions using an Arduino-controlled stepper motor.

The system follows a modular architecture consisting of input, processing, prediction, control, actuation, and feedback modules. This design ensures real-time performance, scalability, and efficient integration of both software and hardware components.

The architecture of the system is divided into three major phases:

5.1. Input and Preprocessing Layer

This layer serves as the entry point of the system, where raw visual data is received from the environment and transformed into a structured format suitable for deep learning inference and motor control.

5.1.1 System Input

The system accepts the following primary input:

- Camera frames (66 × 200 × 3 RGB images) captured from the front-facing vehicle camera, serving as the visual input to the CNN model.
- Serial angle commands received by the Arduino Mega over the UART serial interface at a baud rate of 9600.
- User-defined configuration parameters such as steps per revolution, gear ratio, deadzone threshold, and angle jump limits.

This multi-modal input mechanism enables the system to function both in training (using recorded frames) and in real-time deployment (live camera feed to Arduino).

5.1.2 Preprocessing Stage

Before the data reaches the neural network or the motor controller, it is passed through a series of preprocessing steps:

- Image Resizing and Normalization: Raw camera frames are resized to 66 × 200 pixels to match the CNN input specification and normalized for stable gradient flow during training.
- Batch Preparation: Input images are batched as tensors of shape [None, 66, 200, 3] using TensorFlow placeholders, allowing flexible batch sizes during inference.
- Serial Parsing: On the Arduino side, incoming serial strings are read until a newline character, then parsed as float values representing the target steering angle.
- Angle Clamping and Safety Filtering: The parsed angle is constrained to the range [-360°, 360°] and filtered through a deadzone (0.5°) and jump threshold (70°) to remove noise and prevent erratic motor behaviour.

5.2. System Core Modules

5.2.1 Convolutional Neural Network (CNN) Steering Model

- The core intelligence of the proposed system lies in the CNN-based steering angle predictor. This module directly maps a raw camera image to a continuous steering angle output using an end-to-end deep learning approach based on the NVIDIA Dave-2 architecture.
- The model is defined in model.py using TensorFlow 1.x compatibility mode and comprises the following layers:

Table 2: CNN Layers

Layer	Type	Parameters	Description
Conv Layer 1	Conv2D + ReLU	5×5, 24 filters, stride 2	First feature extraction layer, detects low-level edges
Conv Layer 2	Conv2D + ReLU	5×5, 36 filters, stride 2	Mid-level feature extraction with downsampling
Conv Layer 3	Conv2D + ReLU	5×5, 48 filters, stride 2	Further spatial compression and feature depth
Conv Layer 4	Conv2D + ReLU	3×3, 64 filters, stride 1	High-level feature capture, no spatial downsampling
Conv Layer 5	Conv2D + ReLU	3×3, 64 filters, stride 1	Deep spatial feature encoding
FC Layer 1	Dense + ReLU + Dropout	1152 → 1164 units	Flatten and expand representation with dropout
FC Layer 2	Dense + ReLU + Dropout	1164 → 100 units	Feature compression
FC Layer 3	Dense + ReLU + Dropout	100 → 50 units	Further dimensionality reduction
FC Layer 4	Dense + ReLU + Dropout	50 → 10 units	Near-output feature mapping

Output Layer	Dense + atan × 2	10 → 1 unit	Continuous steering angle prediction in radians

- The final output layer applies an arctangent (atan) activation scaled by a factor of 2. This constrains the predicted steering angle to the range (-π, +π) and prevents unbounded outputs, making it numerically stable for downstream motor control.
- A keep_prob placeholder is applied across all four fully-connected layers to enable dropout regularization. During training, dropout prevents overfitting by randomly deactivating neurons. During inference, keep_prob is set to 1.0 to use the full network.

5.2.2 Stepper Motor Control Module (Arduino Mega)

The motor control module is implemented in step.ino and runs on the Arduino Mega microcontroller. It receives predicted steering angles over the serial interface and translates them into precise stepper motor movements.

Hardware Configuration

The stepper motor is interfaced with the Arduino Mega using two digital control pins:

- STEP_PIN (Pin 2): Generates step pulses to advance the motor one step per HIGH-LOW transition.
- DIR_PIN (Pin 3): Sets the rotation direction — HIGH for clockwise, LOW for counter-clockwise.

•

Table 3:Motor Configuration

Parameter	**Value**	**Description**
Steps Per Revolution	12800	Total steps for a full 360° rotation (microstepping)
Gear Ratio	3.4	Mechanical gear ratio; multiplied into angle to compensate transmission
Angle jump threshold	70.0°	Maximum allowed angle change per update; prevents erratic jumps
Deadzone	0.5°	Minimum angle delta required to trigger motor movement

Serial Baud Rate	9600	UART communication speed with host system
Step Pulse Width	30 µs	HIGH/LOW duration per step pulse for speed control

The control loop follows these sequential steps on every iteration:

- Read incoming serial data and parse a float target angle.
- Clamp the target angle to [-360°, 360°] as a safety bound.
- Compute the angular difference between the target and current positions.
- Reject updates within the deadzone (< 0.5°) to avoid motor jitter.
- Reject updates exceeding the jump threshold (> 70°) to prevent mechanical shock.
- Multiply the angle difference by the gear ratio and invoke rotateToAngle().
- Update currentAngle to reflect the new position.

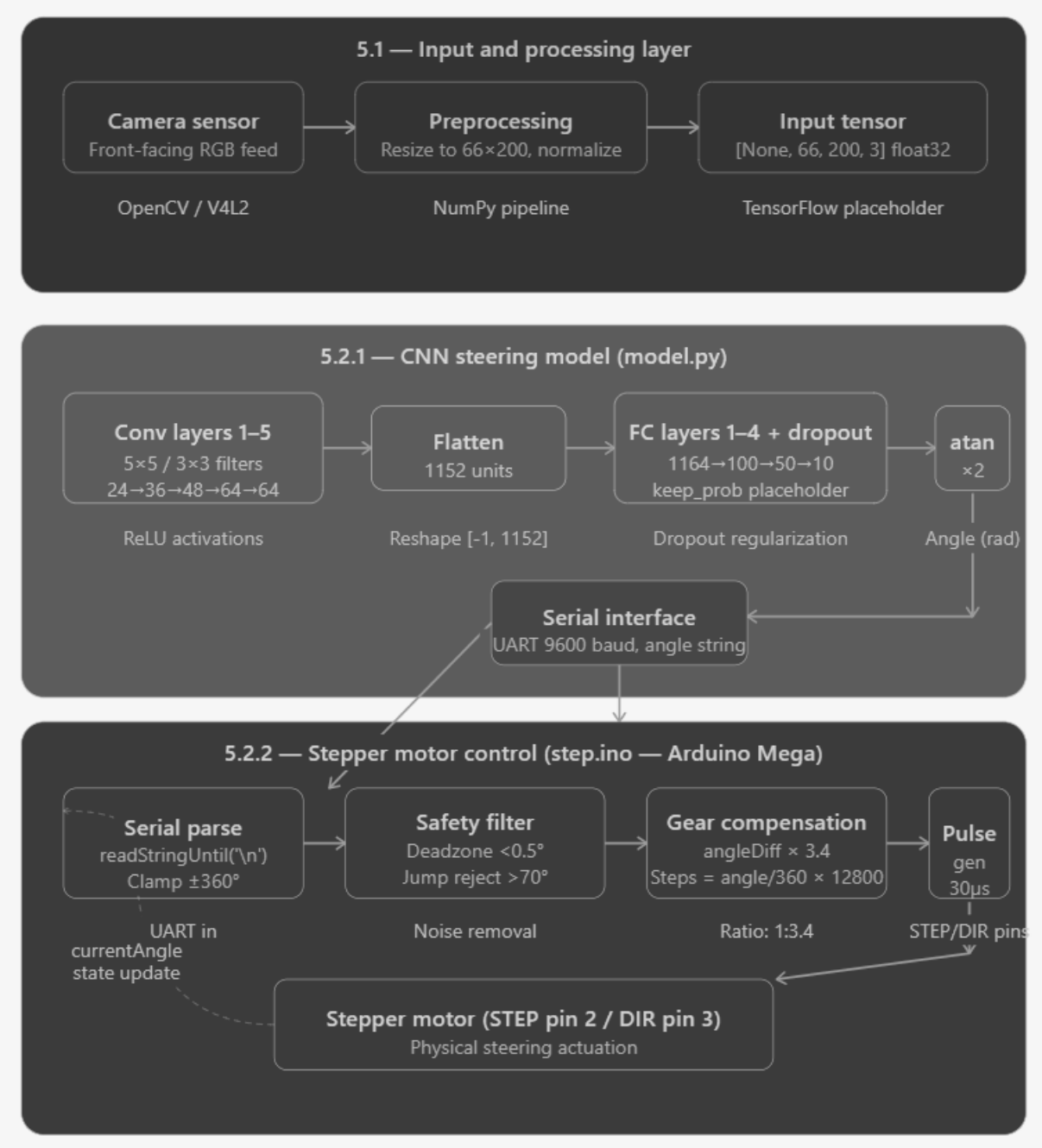

Figure 1: System Architecture Diagram

5.3. Postprocessing and Output Layer

This layer is responsible for converting the raw CNN prediction into actionable motor commands and presenting feedback to the user or monitoring system.

5.3.1 Angle Prediction to Serial Command

Once the CNN produces a steering angle prediction (a floating-point value in radians), the host system performs the following postprocessing:

- The raw radian output is scaled or converted to degrees as required by the serial protocol.
- The angle is serialized as a string followed by a newline character and transmitted via the serial port to the Arduino Mega.
- The transmission occurs at every inference step, keeping the motor synchronized with the model's latest prediction.

5.3.2 Motor Feedback and State Tracking

The Arduino maintains an internal state variable (currentAngle) that tracks the last successfully executed angle. This enables:

- Incremental movement: Only the delta between consecutive predictions is actuated, reducing motor travel and wear.
- Safety enforcement: Consecutive large jumps are detected using the angle history and filtered before reaching the motor.
- Serial monitoring: The Arduino prints rejection messages (e.g., 'Ignored large angle jump.') back to the host over the serial line for real-time debugging.

5.3.3 Result Delivery

The final outputs of the system are delivered in the following forms:

- Physical steering actuation: The stepper motor rotates to the predicted steering angle, directly controlling the mechanical steering mechanism.
- Serial log output: Status messages and rejected-command notifications are streamed back to the connected host terminal.
- Model inference metrics: During development and testing, prediction values and timing metrics can be logged to file or displayed on the development console.

Chapter 6

Experimental Setup

6. Experimental Setup

6.1. Hardware Configuration

The system was developed and tested on the following hardware setup:

- Processor: Intel i5/i7
- RAM: 8 GB / 16 GB
- Storage: 256 GB SSD or higher
- GPU: NVIDIA GPU RTX 3050(for faster LLM inference)
- Arduino Uno R3 (ATmega328P)
- Stepper Motor (NEMA 17)
- Motor Driver (A4988 / L298N)
- Power Supply (12V DC, 2A)
- IoT Module (ESP8266 / ESP32)

6.2. Software Environment

The software setup is designed to support machine learning model development, data processing, and communication with hardware components. It provides an integrated environment for building, training, and deploying the steering prediction system.

1. Arduino IDE

The Arduino Integrated Development Environment (IDE) is used to write, compile, and upload code to the Arduino Uno. It manages motor control logic, sensor data processing, and serial communication with the Python-based ML system. Libraries such as AccelStepper, Wire, Servo, and SoftwareSerial are used to simplify hardware interfacing and improve control efficiency.

2. Python (Version 3.10 or Higher)

Python serves as the primary programming language for implementing machine learning and control logic. It is used for preprocessing input data, running the CNN model, and generating steering predictions. The pySerial library is used to establish communication between the Python environment and Arduino, enabling real-time data transfer.

3. Visual Studio Code (VS Code)

VS Code is used as the main development environment for coding, debugging, and testing. It provides an interactive interface for managing Python scripts, training models, and visualizing results. Its extension support enhances productivity and simplifies project development.

4. Machine Learning Frameworks

TensorFlow / Keras: Used for designing and training the CNN model for steering angle prediction.

PyTorch (Optional): Used for experimentation and alternative model development.

The model is trained using video data converted into frames, enabling it to learn the mapping between road visuals and steering angles.

5. Data Processing and Visualization Libraries

NumPy & Pandas: Used for data handling and preprocessing.

OpenCV: Used for video frame extraction, image processing, and enhancement.

Matplotlib & Seaborn: Used to visualize training performance, accuracy, and loss graphs.

These tools help in improving model performance and understanding system behavior.

6. Version Control (Git & GitHub)

Git and GitHub are used for version control, collaboration, and project management. They allow tracking of code changes, backup of project files, and teamwork during development.

7. Operating System

The system is developed and tested on platforms such as Windows 11 and Ubuntu 20.04. These operating systems provide compatibility with Arduino tools, Python libraries.

Chapter 7

Results and Discussion

7. Results and Discussion

7.1 Steering Angle Prediction Module (CNN Model)

The core functionality of the system is achieved through a Convolutional Neural Network (CNN) that predicts steering angles from input images. The model is trained on a dataset consisting of road images and corresponding steering angles. Each input frame is preprocessed by cropping the lower portion of the image, resizing it to 200 × 66, and normalizing pixel values.

The model outputs a continuous steering angle (in radians), which is converted to degrees using:

$$\boldsymbol{\theta_{deg} = \theta_{rad} \times \frac{180}{\pi}}$$

Where:

- θ_{rad}: Predicted angle in radians
- θ_{deg}: Steering angle in degrees

The predicted angles are transmitted in real-time to the Arduino-controlled steering mechanism. The results show that the model is capable of producing smooth and responsive steering

predictions, especially in straight and moderately curved road conditions. However, occasional noise and sudden spikes were observed, which were mitigated using filtering techniques in the control system.

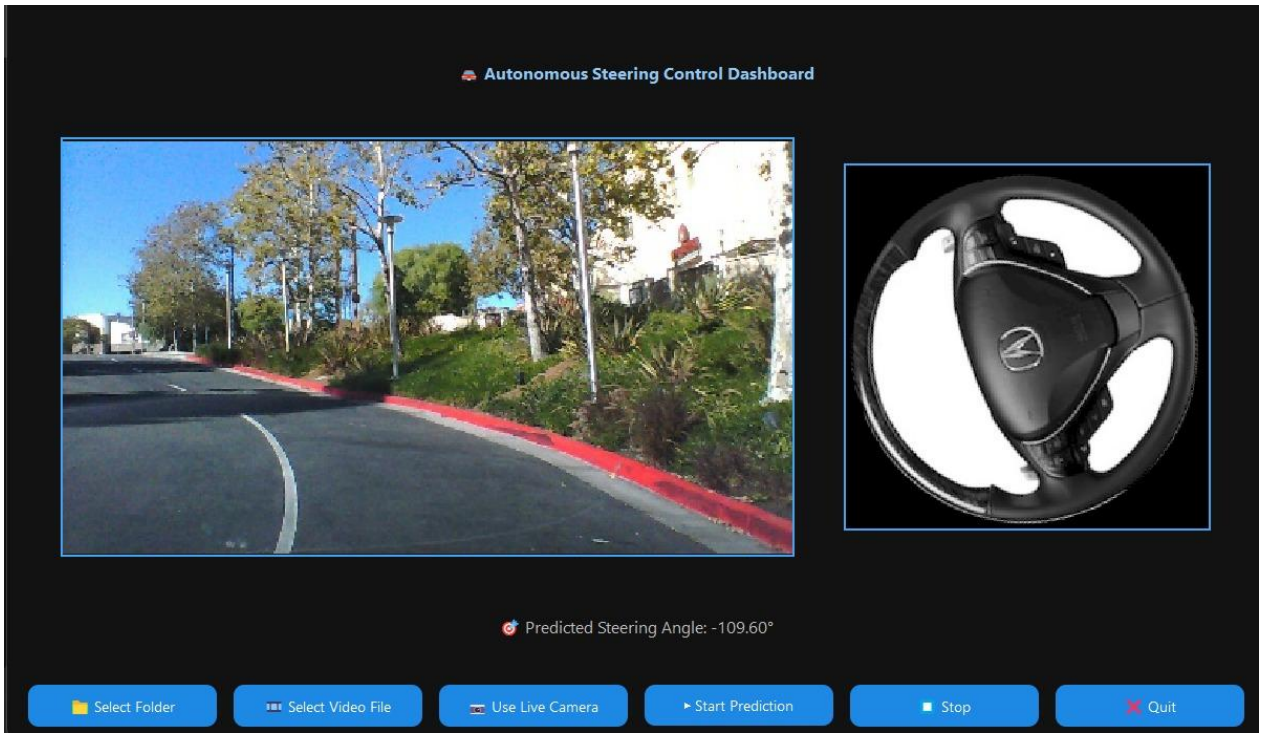

Figure 2: Autonomous Steering Control Dashboard

```
Predicted steering angle: 1.4712322 degrees
Predicted steering angle: 1.7883846 degrees
Predicted steering angle: 3.2915251 degrees
Predicted steering angle: 3.880163 degrees
Predicted steering angle: 5.739716 degrees
Predicted steering angle: 1.140965 degrees
Predicted steering angle: -0.121770486 degrees
Predicted steering angle: 0.5423161 degrees
Predicted steering angle: -2.249536 degrees
Predicted steering angle: -1.9543613 degrees
Predicted steering angle: 5.5948772 degrees
Predicted steering angle: -0.5574664 degrees
Predicted steering angle: -2.5402613 degrees
Predicted steering angle: -4.734021 degrees
```

Figure 3: Predicted Steering Angles per frame

7.2 Steering Control System

The steering control system consists of an Arduino microcontroller, a stepper motor, and a motor driver. The predicted steering angle is sent from the Python script to the Arduino via serial communication.

The number of steps required for rotation is calculated as:

$$Steps = \left| \frac{\Delta\theta}{360} \times N_{steps} \right|$$

Where:

- $\Delta\theta$: Change in steering angle
- N_{steps}: Steps per revolution (12800 in this case)

To ensure safe and stable operation, the system incorporates:

- Deadzone filtering to ignore small fluctuations
- Angle jump thresholding to reject noisy predictions
- Gear ratio compensation for accurate steering control

The results demonstrate that the motor responds accurately to input angles, producing smooth rotational motion. The holding torque observed when current flows confirms proper driver operation.

Figure 4:Real-time Prediction using Stepper Motor

7.3 Noise Reduction and Signal Filtering

Due to fluctuations in CNN predictions, a filtering mechanism was implemented in Python before transmitting commands to the Arduino. A moving average filter using a sliding window was applied:

$$\theta_{avg} = \frac{1}{k}\sum_{i=1}^{k}\theta_i$$

Where:

- k: Window size (5)
- θ_i: Individual predicted angles

Additional constraints include:

- Threshold filtering (ignore changes < 2°)
- Spike rejection (ignore jumps > 60°)

This significantly improved system stability by reducing jitter and preventing erratic steering behavior. The filtered output resulted in more consistent and realistic steering actions.

7.4 System Reliability and Safety

To ensure safe operation, multiple safeguards were implemented:

- Angle clamping (±360°)
- Sudden jump rejection
- Deadzone filtering
- Manual override capability (driver can take control)

These features prevent unexpected or dangerous steering actions, making the system suitable for controlled testing environments.

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

8.1. Conclusion

The project "AI-Driven Autonomous Steering Wheel Using IoT and Machine Learning" successfully demonstrates the development of a low-cost and efficient autonomous steering control system. The system integrates computer vision, deep learning, and embedded hardware to replicate human-like steering behavior. By utilizing a Convolutional Neural Network (CNN), the model is able to analyze real-time video input and accurately predict steering angles based on road conditions.

The predicted steering angles are effectively translated into physical motion using an Arduino-controlled stepper motor, thereby bridging the gap between software-based intelligence and hardware execution. The use of affordable components such as Arduino, stepper motors, and IoT modules ensures that the system remains accessible for students and researchers while still maintaining reliable performance.

The implementation of preprocessing techniques and continuous feedback mechanisms improves system accuracy and stability. Additionally, the integration of IoT enables real-time monitoring and data logging, making the system more interactive and scalable. Overall, the

project achieves its objective of creating a functional and cost-effective autonomous steering prototype.

This work highlights the potential of combining machine learning with embedded systems to develop intelligent transportation solutions. It also provides a strong foundation for further research and development in autonomous vehicle technology.

8.2. Future Work

While the current system effectively focuses on steering control, there is significant potential for further enhancements to improve its overall functionality, performance, and applicability in real-world scenarios. One of the most important extensions would be the integration of braking and acceleration systems. By incorporating throttle and brake control alongside steering, the system can evolve into a fully autonomous driving solution. This would enable the vehicle to make complete driving decisions, such as adjusting speed based on road conditions, traffic, and obstacles, thereby increasing safety and operational efficiency.Another key area of improvement lies in advanced object detection and obstacle avoidance. By integrating powerful computer vision models such as YOLO (You Only Look Once) or SSD (Single Shot Detector), the system can gain the ability to detect pedestrians, vehicles, traffic signs, and unexpected obstacles in real time. This enhancement would significantly improve situational awareness and allow the system to make safer navigation decisions, especially in complex and dynamic environments.

Improving sensor fusion is also crucial for building a more reliable and robust system. By combining data from multiple sensors such as LiDAR, ultrasonic sensors, and GPS, the system can achieve a more accurate understanding of its surroundings. Sensor fusion helps compensate for the limitations of individual sensors, ensuring consistent performance even under challenging conditions like low lighting, fog, or heavy traffic.The adoption of edge computing can further enhance real-time performance. Deploying the model on edge devices such as NVIDIA Jetson or Raspberry Pi reduces dependency on cloud processing and minimizes latency. This ensures faster decision-making, which is critical for autonomous driving applications where even small delays can impact safety.

Enhancing model accuracy and generalization is another important direction for future work. Training the system on diverse datasets that include different road types, weather conditions,

and lighting scenarios can improve its adaptability. A more generalized model will perform reliably across various environments, reducing the risk of failure when exposed to unfamiliar conditions.Cloud-based monitoring and analytics can also add significant value to the system. By extending IoT capabilities, the system can support real-time data visualization, predictive maintenance, and remote monitoring through advanced dashboards. This would allow operators to track performance, detect anomalies, and even control the system remotely when necessary.

Furthermore, real-world vehicle implementation is essential to validate the effectiveness of the system. Testing the model on an actual vehicle in real driving conditions helps identify practical challenges that may not appear in simulated environments. This step is crucial for refining the system and ensuring its readiness for deployment.

Finally, focusing on energy efficiency and system optimization can make the solution more sustainable and practical for long-term use. By reducing power consumption and optimizing computational resources, the system can operate more efficiently, which is particularly important for electric and embedded vehicle platforms.

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***Abstract*—Autonomous driving systems rely on precise steering mechanisms to maintain stability and ensure safe vehicle movement. Among the various control components, estimating the steering angle is essential for functions such as lane alignment, trajectory management, and obstacle response. Conventional rule-based and control-theory approaches have been applied for years, but their effectiveness decreases in unpredictable and highly dynamic road conditions. As a result, data-driven techniques, particularly those based on machine learning, have become a preferred solution. This paper explores different steering angle prediction methods, including regression-based models, convolutional neural networks, and reinforcement learning strategies, as well as vision-oriented, sensor-driven, and integrated hybrid frameworks. The study also discusses how IoT-enabled devices such as cameras, inertial measurement units, GPS modules, and on board processor support continuous data acquisition and real-time decision-making. Existing research is analysed with respect to model structures, datasets, hardware implementations, and evaluation criteria. Additionally, ongoing challenges such as computational efficiency, robustness across diverse driving scenarios, sensor inaccuracies, and safety considerations are addressed, followed by potential future directions emphasizing advanced sensor fusion, edge intelligence, and affordable autonomous steering solutions.**

Keywords- Autonomous vehicles, Steering angle prediction, Lateral vehicle control, IoT-enabled perception.

I Introduction

Autonomous driving technology has seen significant advancements over the last decade, mainly propelled by improvements in artificial intelligence, embedded computing, and intelligent sensing systems. A crucial aspect of this technology is lateral automation, which ensures stable steering control and lane alignment, thus enhancing the safety and reliability of vehicle operation. Accurate lateral control allows vehicles to navigate trajectories, avoid road boundaries, and adapt to dynamic environments. As Advanced Driver Assistance Systems (ADAS) move towards greater autonomy, there is an increasing demand for robust steering automation frameworks.

Lateral control utilized rule-based algorithms and classical control theory, relying on explicit lane geometry and vehicle dynamics. While effective in structured environments, these models struggled with real-world variability, such as changes in illumination, occlusions, road wear, and unpredictable traffic. The advent of large-scale datasets, computational enhancements, and sensor-rich platforms has facilitated a transition to data-driven approaches. Techniques such as machine learning, deep neural networks, and reinforcement learning now enable systems to learn steering behaviours from visual and sensor data, enhancing adaptability and decision-making.

Lane perception is fundamental to lateral automation, with recent strides in computer vision yielding various strategies for lane detection, including segmentation-based and parametric approaches. Steering angle prediction models utilize convolutional neural networks and reinforcementlearning to transform perceptive outputs into actionable control signals. The integration of diverse sensing modalities, like cameras, LiDAR, radar, and inertial measurement units, improves environmental awareness through sensor fusion, mitigating individual sensor drawbacks.

Despite advancements, challenges in lateral automation

persist, particularly regarding real-time processing, computational efficiency, robustness across varied conditions, and safety validation. Furthermore, research often isolates perception, control, or sensing, indicating a need for a holistic approach that interlinks these domains. This paper proposes a comprehensive review of lateral automation techniques and supporting technologies for autonomous vehicles, covering the entire perception-to-control pipeline—from lane detection and steering prediction frameworks to deep learning, reinforcement learning techniques, sensor technologies, and sensor fusion strategies. The aim is to consolidate current advancements, recognize strengths and weaknesses, and outline future research directions for scalable and dependable steering automation

II Lane Detection System

The Advanced Driver Assistance Systems (ADAS) and autonomous driving have advanced quickly due to the increasing use of artificial intelligence in the automotive industry. Real-time vehicle positioning, driving trajectory planning, lane-keeping assistance, and adaptive cruise control are all made possible by lane detection, a fundamental perception task [1]. The majority of lane detection systems currently in use rely on conventional machine learning methods, which are frequently restricted to simple classification tasks rather than intricate feature extraction [2]. Current monocular lane detection methods identify lanes on roads with the help of a camera [3]. Conventional lane detection techniques typically follow a four-step process: image pre-processing, feature extraction, line detection and fitting, and post-processing. These techniques rely on manually created low level features such as colour, gradient, and ridge features. In the previous decade, numerous techniques based on deep neural network have been developed with good performance thanks to advances in deep learning algorithms and computing power [4], [5]. As a type of instance segmentation, lane detection requires that individual pixels be linked to unique lane instances. Conditional instance segmentation techniques have been investigated for lane detection and have demonstrated encouraging results in recent studies. However, because lane markings have distinct structural and geometric features, directly applying these strategies to lane detection is still ineffective [1]. Current approaches can be categorized into four groups based on the line shape description strategy: segmentation-based methods, anchor-based methods, row-wise detection methods, and parametric prediction methods [1], [3].

A. *Segmentation Based Methods*

Among the most popular techniques for lane detection, segmentation based approaches have proven to be effective in real world scenarios. Many of these techniques use a post clustering to increase instance level accuracy.[1], [6]. These methods group pixels that correspond to lane markings into distinct lane instances after first classifying each pixel in the **image using binary segmentation. In order to decode the final lane representations, the resulting instance-level segmentation maps are usually refined and then subjected to post-processing procedures [3], [7].** Many of the earlier methods treat lane detection simply as a pixel level classification task, overlooking the fact that lanes are spatially continuous structures with geometric properties that should be considered during detection [3], [6]. To overcome this problem, anchor-based methods and row-wise detection methods are developed. [1], [3]. Jia-Qi Zhang et al. [3] propose HoughLaneNet, a segmentation-based lane detection method that incorporates Discrete Hough Transform (DHT) based feature aggregation to improve robustness under weak visual cues such as occlusion and lane wear. Unlike traditional segmentation approaches that require multi class instance separation, their method only predicts a binary lane map and performs lane instance discrimination directly in the Hough parameter space. Multi-scale features extracted using a ResNet backbone and FPN are transformed into Hough space, where lane features are aggregated and selected. These features are then fused with spatial features to predict individual lanes as sequences of 2D points, resulting in a simpler yet effective one-stage framework.

B *Anchor based Methods*

Xu et al. [8] introduced an anchor-based lane detection approach that takes advantage of the inherent structural properties of lanes. Using vertical anchors, the method predicts point-wise offsets to reconstruct each lane, and its effectiveness was demonstrated on the CurveLanes dataset. Anchor based lane detection methods follow a top to down pipeline and focus on modelling lane geometry by directly regressing relative coordinates along the lane shape. These approaches reduce ambiguity in regions with weak or missing visual cues and improve instance level lane discrimination. Since lane markings are thin and elongated, conventional box shaped anchors used in object detection are not suitable. Instead, vertical line anchors or slender, lane shaped anchors are commonly adopted to better capture lane structure [1], [9]. Anchor based approaches incorporate prior geometric knowledge of lane structure directly into the model, which helps achieve robust predictions even under heavy occlusion. However, their detection accuracy remains sensitive to how finely the anchors are defined [3].

C *Row wise Detection Methods*

Explainability Row wise lane detection methods effectively exploit prior knowledge about lane shape by predicting the lane position independently for each image row. During training, global shape consistency is enforced through constraints applied to the predicted location at each row. By encouraging continuity and coherence across successive rows, these approaches implicitly model lane geometry and produce smooth, structurally consistent lane

predictions [1], [10], [11]. Motivated by the strong balance between accuracy and computational efficiency achieved by row wise lane detection methods, Lizhe Liu [1] adopts a row wise formulation and introduces several novel strategies to address challenges related to instance-level lane discrimination. Lizhe Liu et al.[1] propose CondLaneNet, a top-down approach that detects lane instances before refining their shapes. By using conditional convolution and a row-wise formulation, the method effectively addresses instance-level ambiguity, while an interaction module helps manage complex lane layouts. The approach achieves strong performance across major benchmarks and maintains real-time efficiency in its lightweight form.

D Parametric Prediction Methods

Unlike the previously discussed approaches that predict discrete lane points, parametric prediction methods directly model lanes as continuous curves defined by parametric equations [1]. Tabelini et al.[12] introduced a parametric lane representation in which lanes are modelled as polynomial curves, with the network directly regressing the corresponding curve parameters. Using third-order polynomials, their approach achieved competitive performance, reaching approximately 90% accuracy on the TuSimple dataset. Despite these promising results, parametric prediction methods have generally not outperformed other lane detection approaches in terms of overall accuracy [1], [3]. With respect to the polynomial order used for lane representation, the marginal accuracy difference observed when employing lower-order polynomials highlights the inherent imbalance of the dataset. Notably, reducing the model to first-order polynomials resulted in an accuracy drop of only 0.35 percentage points [12].

TABLE I METHODS FOR LANE DETECTION.

Method / Model	Core Idea	Contribution
CondLaneNet [1]	Conditional convolution with row-wise formulation	Detects lane instances first, then refines shapes
YOLO-based Lane Detection [2]	Object detection approach	Real-time lane localization
HoughLaneNet [3]	Segmentation + Hough Transform	Hough-space instance separation
Hybrid Spatial–Temporal [4]	Spatial and temporal learning	Improves temporal consistency
LVLane [5]	Robust deep learning	Handles adverse conditions
Self-Attention Distillation [6]	Lightweight CNN learning	Efficient compact models
SCNN [7]	Spatial message passing	Captures lane continuity
CurveLane-NAS [8]	Anchor-based + NAS	Lane-sensitive architecture
LaneATT [9]	Attention-guided anchors	Improves anchor interaction
Ultra Fast Lane Detection [10]	Row-wise prediction	Extremely fast
FastDraw [11]	Sequential prediction	Handles rare lane cases
PolyLaneNet [12]	Polynomial regression	Parametric lane modeling

III Steering Angle prediction

Lane keeping is a core lateral control function that automatically steers an autonomous vehicle to remain within lane boundaries, even under varying road conditions and external disturbances. This is achieved by estimating the desired steering angle from the vehicle's lateral dynamics, which serves as the primary input to the lateral controller for executing stable and accurate lane keeping maneuvers [13]. In autonomous vehicle steering control, vehicle motion is primarily governed by two components: lateral and longitudinal dynamics. Steering specifically addresses lateral motion, with the objective of regulating the vehicle's position within the lane and ensuring stable and accurate path following [14]. Steering angle prediction methods for autonomous vehicles can generally be divided into two main categories. The first category relies on explicitly extracting road or lane boundary information and then applying mathematical or statistical models to compute the steering angle. In contrast, the second category is based on imitation learning, where artificial neural networks learn steering behaviour directly from expert demonstrations. In this approach, lane or boundary features are not manually extracted; instead, the model learns the mapping from visual inputs such as single images or sequences of images to steering angle outputs. In both approaches, the predicted steering angle is combined with other lateral control parameters to ensure that the autonomous vehicle remains safely within road boundaries [13].

A *Steering Angle Prediction Using Computer Vision*

In this steering angle prediction approach, the process begins with explicitly extracting lane markings or road boundary coordinates. Image frames are captured in real time using a camera mounted on the vehicle and are then pre-processed to enhance important road features and reduce noise. Based on these refined images, lane boundaries are detected and used to compute the required steering angle. To improve robustness in challenging situations such as occlusions, faded lane markings, or illumination changes, some studies further apply lane tracking across consecutive frames. This temporal consistency is achieved using filtering techniques such as Kalman filters, particle filters, or Bayesian filters, allowing reliable road detection even when lane information is partially missing or noisy [13]. Traditionally, the pre-processing stage relies on a set of image enhancement techniques to simplify the visual input. These commonly include reducing image resolution through down sampling, segmenting the image to isolate relevant features,

smoothing to suppress noise, selecting a Region of Interest that focuses on the road area, and applying Inverse Perspective Mapping (IPM) to obtain a clearer representation of lane geometry [13], [14]. The steering angle is estimated by finding the intersection of the detected left and right lane boundaries. If this point aligns with the road centre then no steering adjustment is needed. Otherwise, a central reference path is formed using the lane geometry and the steering angle is computed from the deviation between this path and the vehicle's current heading [13].

B Steering Angle Prediction using Deep Learning

Deep learning has significantly improved steering angle prediction in autonomous vehicles by enabling models to learn directly from raw, unlabelled data. By analysing the overall scene context and focusing on relevant features at multiple levels of detail, deep learning methods can effectively capture both fine-grained and high-level visual information needed for accurate steering decisions [14]. This section reviews the deep learning architectures used for steering angle prediction in autonomous vehicles. Aires Da Conceicao [15] proposed a steering and road direction prediction system based on the NVIDIA end to end convolutional neural network architecture. The system uses road images captured in real time by a camera mounted on the vehicle as input to learn the relationship between visual features and driving direction. The method follows a supervised learning approach in which labelled image data are used to train a convolutional neural network model. After training, the model predicts the road direction directly from camera images and produces a continuous output that supports steering control in autonomous vehicles. Rodolfo Valiente [16] addressed steering angle prediction as a regression problem by directly mapping visual input to continuous steering values. The study highlights that using a sequence of images rather than a single frame overcome challenges such as strong sunlight, image noise, or dead end roads. Temporal relationships between consecutive frames help the system infer more reliable steering decisions. To model these temporal dependencies, Valiente employed Long Short Term Memory networks, which retain relevant information from previous frames and use it to predict the next steering angle. The proposed approach further extends this idea by incorporating vehicle to vehicle communication. Images captured by a leading vehicle are shared with a following vehicle and combined with locally captured images, allowing the system to anticipate road conditions beyond the current field of view. The model uses an end to end deep learning framework combining CNN, LSTM, and fully connected layers to predict steering angles. CNN extracts visual features, LSTM captures temporal information across frames, and the final layers output a continuous steering value using regression with mean squared error loss [13], [14], [16].

C Steering Angle Prediction using Deep Reinforcement Learning

Deep Reinforcement Learning combines deep learning and reinforcement learning. It has enabled solutions to complex tasks that were difficult for traditional machine learning methods. In this approach an agent learns through trial and error by interacting with its environment and receiving rewards with the goal of maximizing long term returns over time [14]. Qadeer Khan [17] proposed a deep reinforcement learning based steering control framework that combines perception and decision making in an end to end manner. In this approach, images from a front facing camera are first processed by a perception model to generate a compact semantic representation of the driving scene. Rather than using raw images directly, the reinforcement learning agent is trained on this latent semantic embedding, which reduces complexity and improves learning efficiency. The steering controller is trained through interaction with the environment using a reward driven trial and error process instead of supervised labels. Positive rewards are assigned for safe lane following, while negative rewards are given for unsafe behaviours such as lane departure, off road driving, or collisions. Training is carried out in the CARLA simulator, where the agent learns appropriate steering actions by maximizing long term cumulative rewards. To address the large continuous state space, the study adopts a model free Deep Q Learning strategy, where a neural network approximates state action values instead of relying on explicit environment models. Techniques such as experience replay and a separate target network are used to stabilize training. By operating on semantic latent states rather than full images, the method significantly reduces training time while maintaining effective steering control performance. Keyu Wu [18] introduced a deep reinforcement learning based steering control framework called BND-DDQN, designed to model complex nonlinear driving policies. The approach is built on a Double Deep Q Network architecture enhanced with branching, dueling, and noisy components to improve learning stability and exploration efficiency. The model processes a short sequence of depth images using convolutional layers to extract compact feature representations, which are then shared across multiple decision branches. These branches separately estimate the overall state value and the action advantages associated with linear and angular vehicle motions. By separating these components, the network is able to learn more precise control policies for steering and speed. To encourage effective exploration during training, the framework replaces traditional random exploration strategies with noisy neural networks, where controlled noise is injected directly into network parameters. This allows the agent to explore the environment in a more structured and consistent manner. The estimated state values and action advantages are combined to produce final action value predictions, and actions are selected greedily based on the learned policy without relying on epsilon greedy exploration.

TABLE II STEERING ANGLE PREDICTION TECHNIQUES.

Method	**Model / Algorithm**	**Dataset / Platform**	**Key Contribution**
Deep Learning (Imitation	CNN (NVIDIA end to end)	Real world images	End to end steering angle prediction from road images

Method	Model / Algorithm	Dataset / Platform	Key Contribution
Learning) [15]			
Deep Learning (CNN + LSTM) [16]	CNN + LSTM + FC	Simulation and V2V	Temporal steering prediction using sequential images and cooperative perception
Deep Reinforcement Learning [17]	DQN based RL	CARLA simulator	Latent space reinforcement learning for steering control
Deep Reinforcement Learning [18]	BND DDQN	Simulation	Branching dueling noisy DDQN for stable nonlinear steering control

IV Sensor Technology for Lateral Automation

Sensors are essential components of autonomous vehicles as they convert physical changes in the vehicle or its surroundings into numerical data that can be processed by perception and control systems. Based on their operational principle sensors are generally classified into internal state and external state sensors. Internal state sensors also known as proprioceptive sensors monitor the dynamic condition of the vehicle by measuring variables such as angular rate wheel speed force and battery status. Typical examples include inertial measurement units encoders gyroscopes magnetometers and global navigation satellite system receivers. In contrast external state sensors referred to as exteroceptive sensors capture information from the surrounding environment including object distance road features and light intensity. Common exteroceptive sensors used in autonomous vehicles include cameras radar LiDAR and ultrasonic sensors. Sensors may further be categorized as active or passive depending on whether they emit energy into the environment. In autonomous driving applications the integration of these sensors is essential for environment perception obstacle detection and safe path planning [19], [20].

A Cameras

Camera based vision sensors operate on optical imaging principles in which incoming light is focused through lenses onto image sensors such as CCD or CMOS. CCD sensors transfer charge sequentially which results in high image quality but higher power consumption. In contrast CMOS sensors integrate signal amplification and readout within each pixel enabling higher resolution images with rich colour texture and shape information. As a result camera sensors are widely used for semantic perception tasks including lane detection traffic sign recognition and pedestrian classification. Cameras are compact and cost effective which makes them attractive for large scale deployment. However they lack direct depth perception and their performance is sensitive to poor ligh zting conditions shadows and occlusions [21]. Stereo cameras, often called binocular cameras, replicate biological depth perception by using the difference between two slightly offset images to infer distance. Each system consists of two image sensors separated by a fixed baseline, which is the physical distance between the sensors and varies by camera design. Orbbec 3D cameras commonly used in autonomous and intelligent vehicle applications employ a baseline of about 75 mm. By analysing image disparity between the paired views, stereo cameras can generate depth information through geometric triangulation. This depth output is widely used in robotics, where stereo vision pipelines, such as those available in the Robot Operating System, support tasks including simultaneous localization and mapping and autonomous navigation [20].

B LiDAR

A key perception sensor for contemporary Advanced Driver Assistance Systems and autonomous cars, LiDAR (Light Detection and Ranging) is a remote sensing technology that began with aerial and terrain mapping applications. The technology allows for accurate distance estimation by generating laser pulses and timing the return of reflected signals from nearby objects. LiDAR continuously scans the environment to create a point cloud, a 3D representation of the scene that helps autonomous driving systems perceive the environment accurately and comprehend space [20]. Camera sensors are affordable but they do not provide accurate depth information which limits their effectiveness for precise three dimensional object detection. Their performance also degrades in challenging weather conditions such as fog rain and snow. Sensors that produce point cloud data such as LiDAR and radar overcome these limitations by directly measuring depth which enables more reliable estimation of object size and position. However point cloud data are sparse irregular and unevenly distributed which makes them unsuitable for direct use with conventional image based deep learning methods. As a result specialized processing techniques are required to handle this type of data and to reduce sparsity related issues. In autonomous driving research point cloud representations are generally grouped into projection based voxel based and raw point cloud approaches each offering different trade-offs between accuracy and computational efficiency [22].

C Radar

Radar is a long established sensing technology that detects surrounding objects by transmitting electromagnetic waves and analysing their reflections. The time taken for the signals to return is used to estimate the distance to targets while changes in the signal frequency caused by relative motion enable the estimation of object

speed. This combined ability to measure range and velocity makes radar well suited for identifying and tracking moving obstacles in automotive environments [20], [21]. Radar datasets used in autonomous driving research differ significantly in terms of sensor type and data representation. Radar sensors are commonly categorized into low resolution and high resolution systems, with high resolution achieved through various technical approaches such as advanced antenna configurations, cooperative sensing, and synthetic or extended aperture techniques. Commercial automotive radars typically output point cloud data containing range azimuth Doppler velocity and radar cross section information, while newer four dimensional radars additionally provide elevation measurements. Some experimental radar systems also offer access to raw radar data including digitized signals and intermediate representations. In autonomous driving applications radar is mainly employed for localization and object detection. While detection remains the primary focus in many studies localization datasets are also relevant since they often include synchronized radar LiDAR and camera data which can be repurposed for detection tasks. Radar annotations are provided at different levels of detail depending on the dataset. These range from two dimensional bounding boxes in bird's eye view with orientation information to full three dimensional bounding boxes that additionally encode object height and orientation [23].

D Inertial Measurement Unit

Inertial navigation also called inertial odometry estimates the position orientation and three dimensional velocity of a moving system using data from an Inertial Measurement Unit. The IMU contains accelerometers and gyroscopes and enables real time motion estimation through dead reckoning without relying on external references. While inertial navigation has been studied for many years the availability of small and low cost sensors has increased its use in robotics and autonomous systems. High precision inertial navigation systems can provide very accurate localization but they are expensive and require complex initialization. In contrast low cost IMUs are more practical for autonomous platforms but suffer from significant errors such as sensor noise bias and drift. These errors accumulate over time and lead to rapid degradation in localization accuracy. As a result inertial navigation alone is not sufficient for long term autonomy and is commonly combined with other sensors such as GNSS cameras or LiDAR to improve reliability and accuracy [24] [19]. Martin Brossard [24] presents a long-term localization approach for wheeled robots that relies solely on inertial measurements. The method takes advantage of characteristic motion patterns commonly observed in wheeled vehicles, which provide valuable information about the system's motion when correctly identified. These motion patterns are detected automatically in real time using only IMU data through a learning-based model trained on inertial measurements and ground truth trajectories. The detected motion information is integrated into an advanced Kalman filtering framework, where it is treated as additional constraints and statistically combined with raw IMU measurements. This enables accurate estimation of vehicle pose, velocity, and sensor biases, along with their associated uncertainty. Experimental results on a public vehicle dataset show that the proposed IMU-only approach achieves high long-term accuracy over extended driving sequences. The method outperforms conventional wheel odometry as well as wheel odometry aided by high-grade gyroscopes with significantly lower drift. These findings demonstrate that identifying meaningful patterns in inertial data and incorporating them into a principled filtering framework can significantly enhance localization. Moreover, the approach is flexible and can be extended to include measurements from other sensors, making it a robust building block for broader autonomous localization systems.

E Sensor Fusion

Sensor fusion is a fundamental component of autonomous systems such as self-driving cars and unmanned ground vehicles. It combines data from multiple sensors to reduce uncertainties and address the limitations of individual devices while providing a more accurate and consistent understanding of the environment. By integrating complementary modalities such as camera and radar autonomous systems can obtain both high resolution visual information and measurements of obstacle velocities which improves situational awareness and supports reliable navigation in diverse conditions [20].

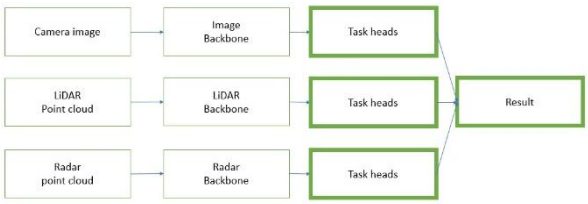

Figure 1: Late Sensor Fusion Flow

Fusion in autonomous driving systems are divided into three stages: early fusion, mid-level fusion, and late fusion. Each stage reflects the timing and abstraction level at which sensor data from cameras, LiDAR, Radar, and sometimes IMU or GPS, is integrated within the processing pipeline, Early fusion combines raw data from multiple sensors before feature extraction allowing the system to learn joint representations from the start but it requires large models and can be affected by differences in sensor resolution coordinate frames sampling rates and noise. Mid-level fusion extracts meaningful features from each sensor independently before combining them which reduces input size mitigates alignment issues and can handle missing or degraded data. However it may lose some fine details across sensors. Late fusion merges outputs at the decision level providing robustness modularity and computational efficiency but it is less effective for tasks that require detailed spatial or semantic integration [25]. The figure 1 shows the flow in the late fusion.

V Conclusion

Autonomous steering control has become a fundamental component in the progress of self-driving vehicle technology, particularly with the integration of machine

learning and IoT-based systems. The review of existing research shows a clear transition from conventional rule-based and control-theoretic methods to data-driven approaches capable of learning steering behaviour directly from visual and sensor inputs. Techniques such as convolutional neural networks have demonstrated strong capability in predicting steering angles with reasonable accuracy, while embedded platforms like Arduino and related IoT modules enable practical real-time actuation at a lower cost. Although high-end systems equipped with advanced sensors achieve high precision, their complexity and expense limit accessibility for academic and small-scale experimentation. Low-cost implementations illustrate that effective steering automation can still be achieved using optimized models and modular hardware integration. However, challenges related to real-time inference, robustness under varying environmental conditions, sensor noise, and operational safety continue to require careful attention. Overall, the combination of efficient learning algorithms and scalable IoT hardware presents a promising pathway toward affordable, adaptable, and research-oriented autonomous steering solutions.

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